Where the Shortcut Sits

Status report 1: does masking the region of interest remove artifact shortcuts?

October 2026

Summary.

- Medical image classifiers often learn *artifacts* (rulers, hair, calipers, debris) instead of the disease. A common approach is *ROI masking*: keep only the region of interest (ROI), such as the lesion, and delete the rest of the image.
- Masking removes the artifact only when the artifact lies *outside* the ROI. When it lies *inside*, masking keeps it and removes only the background outside the ROI; in most of our tests the model then relied on the artifact *even more*.
- We showed this in four steps: (1) moving a pasted ruler into a skin lesion step by step; (2) four checks of *why* it happens; (3) the same test in four other kinds of medical images; (4) real, unedited artifacts in four patient cohorts.
- With real artifacts, masking helped much more when the artifact was outside the ROI than when it was inside, in all four cohorts (all four planned tests significant after correction for multiple testing; Section 2). Two image encoders were exceptions (Section 7).

1 The problem

A deep classifier learns whatever predicts the label in its training data, including things that are not the disease (Geirhos et al., 2020). This is called a *shortcut*. In medical images the shortcut is often an artifact that clinicians put there because they suspected disease: a ruler or skin marking next to a suspicious mole (Winkler et al., 2019, 2021), calipers on a thyroid nodule that was measured, or a chest drain in a patient with a collapsed lung (Jiménez-Sánchez et al., 2023). Because these artifacts appear more often in sick patients, a model can score well by detecting the artifact. It then fails on exactly the patients whose images look different, for example a melanoma photographed without a ruler.

A common approach is to segment the lesion and give the model only that region; the rest of the image is deleted. This is called *region-of-interest (ROI) masking*. It is popular because it needs only a lesion mask, which many pipelines already compute. Its hidden assumption is that the artifact lies in the background. Rulers, hair, calipers and debris often lie on or across the lesion. Figure 1 shows what masking then does.

Research question. *Does the benefit of ROI masking depend on where the artifact sits relative to the ROI? If so, why, and does it hold for real artifacts and other kinds of medical images?*

2 How we test it

Two models Every experiment compares two models that use the same frozen image encoder (a pretrained foundation model; mainly DINOv2 ViT-B/14 at 518 px (Oquab et al., 2024)) and the same simple classifier on top (logistic regression whose settings are chosen on clean validation images):

- the **standard model** sees the whole image;
- the **masked model** sees the image with everything outside the ROI replaced by the image's average colour, in training and in testing.

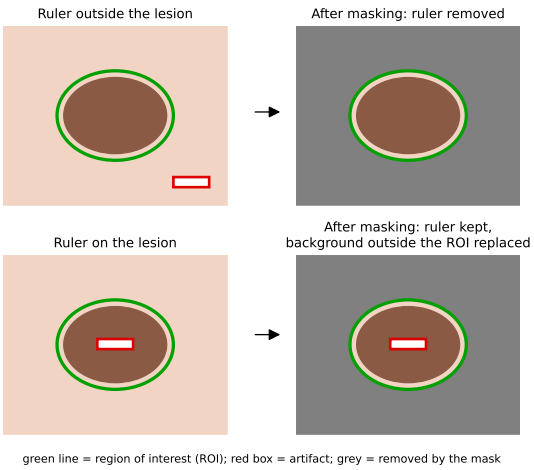


Figure 1: The idea. Top: an artifact outside the lesion is deleted by the mask. Bottom: an artifact on the lesion survives the mask, while the background around the lesion is replaced.

The two models share the frozen encoder, the classifier type and the training recipe; only the input differs. Each classifier is trained with 3 random seeds in the pasted-artifact experiments. In the real-artifact experiments it is trained with 5 seeds, and for each seed on 5 grouped folds (25 fits per model). The folds are grouped by lesion for skin (ISIC lesion IDs, plus near-identical images) and by patient for chest X-rays (NIH patient IDs; no patient is in two splits). Our copies of the thyroid, ovarian and capsule datasets carry no patient identifiers, so there we group near-identical images (and, for capsule images, blocks of consecutive frames).

The trap To measure how much a model relies on the artifact, we build a "trap" (Figure 2). In the training data the artifact appears on 90% of the images of the target class (e.g. melanoma) and 10% of the other class, so it is a strong shortcut. We then test on three test sets:

- **correlated test**: same 90/10 pattern as in training. A shortcut model looks good here.
- **reversed test**: the pattern is flipped (10/90). A model that uses the artifact *fails* here.
- **clean test**: the artifact tells nothing about the disease (pasted-artifact experiments: no artifact; real-artifact experiments: artifact on half of each class). This shows how well the model knows the disease itself.

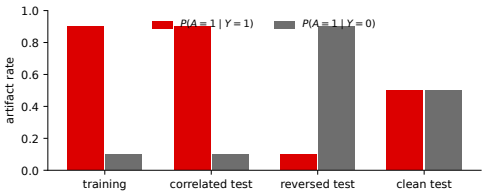


Figure 2: The trap. Bars show how often the artifact appears in the target class (red) and the other class (grey) in training and in each test set. The clean test shown (50/50) is the one of the real-artifact experiments; in the pasted-artifact experiments the clean test has no artifact.

The score We report AUROC (0.5 = guessing, 1.0 = perfect). Our main score is the AUROC on the *reversed* test. A higher reversed-test AUROC means better discrimination when the artifact–label link is reversed. It does not directly measure reliance on the artifact: a model can rely on the artifact and still score well if it also knows the disease well. We usually report the *difference* "masked minus standard". A positive number means masking helped; a negative number means it hurt. Square brackets give the 95% confidence interval (10,000 bootstrap replicates; see "Honest error bars"); if it does not include zero, the difference is significant. The **location effect** is this difference with the artifact outside the

ROI minus the same difference with the artifact fully inside. A positive location effect means masking helps more when the artifact is outside.

Measuring fairly Two details turned out to matter.

- **Near-duplicate images.** Skin-lesion datasets contain several photos of the same lesion. If one copy is in training and another in testing, the test partly measures memory. We always split by lesion (and by image similarity), never by single images. With random splits, 93 near-duplicate pairs crossed the split and the measured shortcut was 10–18% too large (shortcut gap 0.284 vs 0.241 with the ruler outside, 0.465 vs 0.421 inside).
- **Honest error bars.** We train each model several times, and the runs share test images. In the pasted-artifact experiments, all three seeds score one fixed test set (356 images, 73 melanomas; training 1,735, validation 346). In the real-artifact experiments, each of the 5 seeds tests every image once over its 5 folds, so the seeds' test sets overlap heavily. A common way of computing confidence intervals treats every run as if it had its own test set, which makes the intervals too narrow. Instead, each bootstrap replicate resamples the seeds and gives each test image one random weight that it keeps in every seed, in both models and (for the crossover) in both traps; the compared numbers are therefore always paired on the same images. Our intervals are about 1.5 times wider than the common method's, and a few results that looked significant with it are not. With one bootstrap weight per lesion or near-duplicate group instead of per image, the intervals of the four planned tests change in width by a factor of 0.97–1.50 (most for capsule frames) and all four tests stay significant (Holm-adjusted $p \leq 0.0008$).

Before each main experiment we wrote down the design and what result would count as support, and stored it in the project repository. Results are reported whichever way they went.

3 Step 1: moving a ruler into the lesion

What we did We first wanted to change *only* the location of the artifact. We took 2,437 dermoscopy images with manual lesion masks (ISIC 2018 (Codella et al., 2019); task: melanoma or not) and pasted the same ruler into each image at 0, 25, 50, 75 or 100% overlap with the lesion (Figure 3). Only the ruler's position changes: same images, same ruler, same labels.

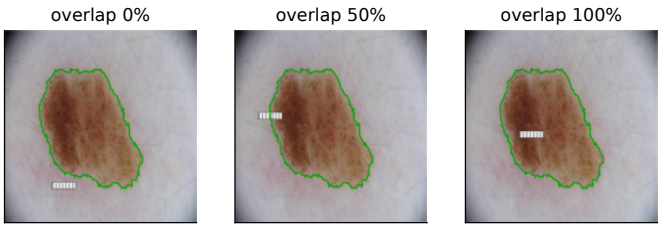


Figure 3: The same image with the ruler at 0, 50 and 100% overlap with the lesion (green: lesion mask).

What we found When the ruler lies outside the lesion, masking removes it and helps a lot (+0.18). From 50% overlap on, masking *hurts*: at full overlap the masked model is worse than the standard model (−0.13; Table 1, Figure 4). The correlated test shows why: with the ruler on the lesion, the masked model scores *higher* than the standard model there (0.961 vs 0.937) but lower on the reversed test (0.386 vs 0.515). It has become *more* dependent on the ruler.

Table 1: Ruler experiment (DINOv2, AUROC, mean of 3 seeds). The last row is the effect of masking on the reversed test.

	Ruler outside (0%)	Half on the lesion (50%)	Fully on the lesion (100%)
Standard model, reversed test	0.665	0.573	0.515
Masked model, reversed test	0.849	0.476	0.386
Standard model, correlated test	0.906	—	0.937
Masked model, correlated test	0.849	—	0.961
Masked − standard (reversed)	**+0.184** [+0.117, +0.250]	**−0.097** [−0.164, −0.030]	**−0.129** [−0.195, −0.064]

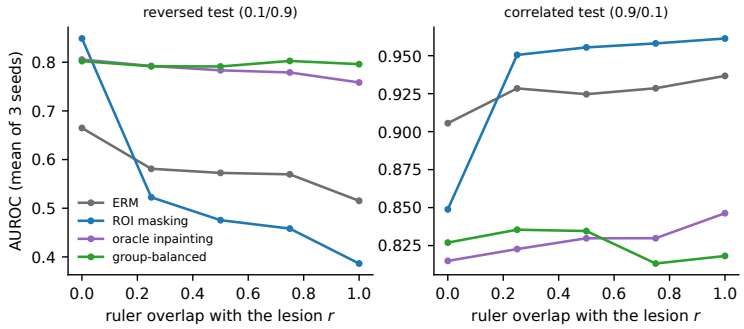


Figure 4: AUROC on the reversed test (left) and the correlated test (right) as the ruler moves into the lesion. Blue: masked model; grey: standard model. For comparison: purple, painting out the ruler (needs the ruler's outline); green, balanced reweighting (needs to know which images have a ruler).

Is it a fluke of one model or one ruler? No. We repeated the experiment with a lower image resolution (224 px), with a dermatology-specific encoder (DermLIP (Yan et al., 2025)), and with a ruler whose colour, ticks and size change from image to image. The location effect was large and positive every time (Table 2, first four rows).

Table 2: The ruler result repeated with other settings (masked − standard, reversed test).

Setting	Ruler outside	Ruler fully inside	Location effect
DINOv2, 518-px images (main result)	+0.184	−0.129	+0.313 [+0.258, +0.370]
DINOv2, 224-px images	+0.361	−0.095	+0.456 [+0.390, +0.525]
DermLIP (dermatology encoder)	+0.161	−0.263	+0.424 [+0.335, +0.517]
DINOv2, randomised ruler	+0.146	−0.155	+0.302 [+0.248, +0.357]
Two ruler sizes on the same 1,314 images:			
DINOv2, ruler 13% of the image width	+0.160	−0.102	+0.262 [+0.200, +0.325]
DINOv2, ruler 39% of the image width	+0.170	−0.054	+0.224 [+0.161, +0.285]
Average of the two sizes	**+0.165**	**−0.078**	**+0.243** [+0.182, +0.304]

Different ruler sizes Real rulers come in different sizes. We therefore ran two sizes, 13% and 39% of the image width (the wider one thin, with millimetre ticks, like a real scale bar), on the 1,314 images whose lesions can hold both fully inside (last three rows of Table 2). The design was written down before the run; the average was added afterwards. Both sizes show a clear location effect (+0.262 and +0.224). Averaged over the two, masking helps by +0.165 with the ruler outside and hurts by −0.078 [−0.149, −0.010] with it fully on the lesion. For the 39% ruler alone, that harm is not significant (−0.054; the interval includes zero). With both sizes, the masked model again reacts to the ruler on the lesion about twice as strongly as the standard model (check 3 below).

4 Step 2: why does masking fail?

Step 1 showed *that* masking fails when the ruler is inside. There were three possible reasons: (a) masking simply hides useful parts of the image; (b) the background holds disease information that masking throws away; (c) the ruler inside the lesion survives, and because masking deletes everything that competed with it, the model relies on it more. We designed one check for each (Table 3).

Table 3: Four checks of why masking fails (ruler experiment, DINOv2).

Check	What we changed	Result	What it means
1. Useless ruler	Ruler at the same places, but *unrelated* to the diagnosis	No statistically clear harm at any position (all intervals include 0; e.g. +0.039 at 100%)	Supports dependence on the ruler as a shortcut; does not exclude an effect of hiding image parts (speaks against a as the main cause)
2. Bigger mask	Grow the mask around the lesion: more skin comes back, and more of a half-covered ruler stays visible (50% → 95% of the ruler)	Harm gets *worse*: $-0.097 \rightarrow -0.153$. When the ruler is already fully visible, the point estimates were similar (-0.129 vs -0.138)	At 50% skin and ruler come back together; at 100% only skin comes back and the estimates were similar. Speaks against b as the main cause
3. Same-image test	Add or remove the ruler on the *same* test image and measure how much the predicted probability (0 to 1) changes	The masked model's prediction changes about twice as much as the standard model's (0.49 vs 0.24; with the 39% ruler 0.43 vs 0.21)	The masked model leans more on the ruler (supports c)
4. Lesion size	Split test images into small, medium and large lesions	Harm with the ruler inside: small -0.36, medium -0.16, large $+0.06$ (not significant)	Harm is largest where the ruler is big compared to the lesion

Explanation. Masking deletes everything outside the ROI. If the artifact is outside, the shortcut is gone. If the artifact is inside, it survives, and the other image information that used to compete with it is gone, so the artifact becomes a bigger part of what the model can use. The model relies on it more, and fails when the artifact is missing or misleading.

5 Step 3: is it only skin images?

What we did We repeated the moving-artifact experiment in four other kinds of medical images, each with its own typical artifact and its own ROI (Figure 5): a sonographer's caliper on **thyroid ultrasound** (TN3K, malignant or benign nodule (Gong et al., 2021)) and on **ovarian ultrasound** (MMOTU (Zhao et al., 2022); our grouping of its diagnoses, normal ovaries excluded; tumour with a solid part: teratoma, theca cell tumour, mucinous cystadenoma, high-grade serous carcinoma; cystic: chocolate cyst, serous cystadenoma, simple cyst); a debris blob on **capsule endoscopy** (SEE-AI, erosion or polyp-like lesion (Yokote et al., 2023)); and a tube on **chest X-rays** (NIH ChestX-ray14, pneumothorax (Wang et al., 2017); ROI = the lungs). We used only images without a real artifact, so the pasted one was the only one. We also tested a second encoder where possible (MedSigLIP, a medical image model; RAD-DINO, a chest X-ray model (Pérez-García et al., 2025)).

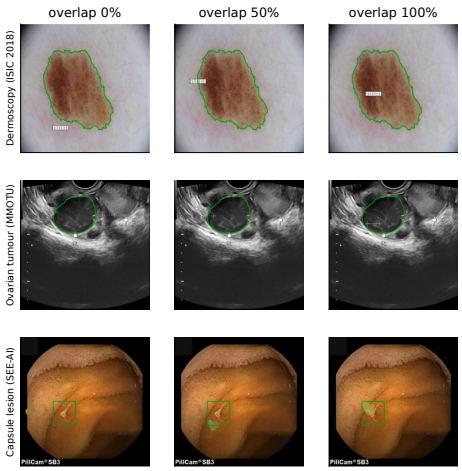


Figure 5: The pasted artifact at 0, 50 and 100% overlap in three modalities (green: ROI; for capsule images the ROI is the expert's box around the lesion).

What we found Counting the first four rows of Table 2 and the seven rows of Table 4, the location effect was positive in 10 of 11 experiments (Figure 6). Masking helped clearly when the artifact was outside and lost most or all of its benefit when it was inside. Whether masking actually *hurt* at full overlap depended on the setting: it hurt significantly in 7 of the 11. In thyroid ultrasound, for example, the benefit fell from +0.40 to about zero.

Table 4: Moving-artifact experiments in other modalities (masked − standard, reversed test).

Modality and artifact	Encoder	Artifact outside	Artifact fully inside	Location effect
Thyroid ultrasound, caliper	DINOv2	+0.402	−0.030	+0.433 [+0.362, +0.507]
	MedSigLIP	+0.514	+0.044	+0.470 [+0.388, +0.556]
Ovarian ultrasound, caliper	DINOv2	+0.041	−0.197	+0.238 [+0.144, +0.340]
Capsule endoscopy, debris	DINOv2	+0.116	−0.112	+0.228 [+0.144, +0.324]
	MedSigLIP	+0.374	+0.216	+0.158 [+0.012, +0.314]
Chest X-ray, tube	DINOv2	+0.254	−0.083	+0.337 [+0.312, +0.364]
	RAD-DINO	−0.002	+0.111	−0.113 [−0.129, −0.097]

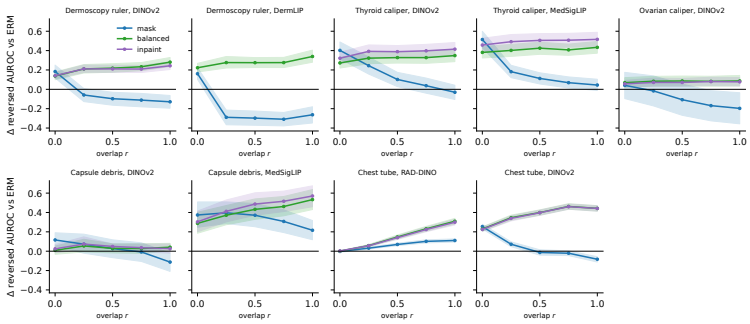


Figure 6: Effect on the reversed test (vs the standard model) as the artifact moves into the ROI, in nine of the eleven moving-artifact experiments (the 224-px and randomised-ruler runs are in Table 2 only). Blue: masking; purple: painting out the artifact; green: balanced reweighting. The blue line falls in every panel except the RAD-DINO chest tube.

The one exception With the chest X-ray encoder RAD-DINO, the standard model ignored a tube outside the lungs (reversed AUROC 0.913, the same as on the clean test, 0.915), so masking gained nothing there, while it helped with the tube inside the lungs (+0.111). This is a real exception; the likely reason is that this encoder does not use a tube outside the lungs. With DINOv2, which did use it, the usual pattern appeared.

6 Step 4: real artifacts

Pasted artifacts let us control location exactly, but they are not the artifacts clinicians produce. So we next asked whether the same thing happens with *real, unedited* artifacts.

The design A real artifact cannot be moved, so we compare two groups of images: those whose artifact lies mostly inside the ROI (**Trap A**) and those whose artifact lies outside it (**Trap B**). Both traps share the *same* artifact-free images (Figure 7). The two groups of artifact images can still differ in other ways (lesion size, diagnosis mix, hospital, the artifact's appearance); we matched the hospital mix where there were several hospitals, but the Trap A versus Trap B comparison remains an association with location, not a controlled move of the artifact. Each trap gets the 90/10 training set and the three test sets of Figure 2. The **crossover** is the benefit of masking in Trap B minus its benefit in Trap A; a positive crossover means masking helps more when the artifact is outside. (We first tried comparing in-ROI with out-of-ROI images directly, without an artifact-free group. That design cannot work: masking deletes exactly the out-of-ROI artifacts that define one side of the comparison, so it shows "harm" whatever the truth.)

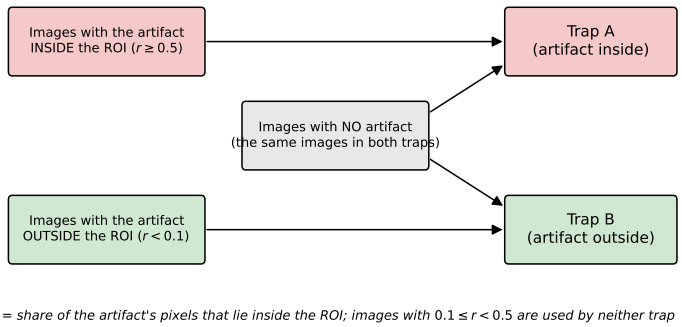


Figure 7: Two traps built from real images. Both share one group of artifact-free images.

Cohorts Table 5. Artifacts were located automatically: for skin images, published masks that mark hair and rulers together (Wegley et al., 2026), so this artifact is called "hair/ruler"; a rule-based caliper detector for ultrasound; and a small detector trained on expert debris masks for capsule images. We checked them by eye: the caliper detector was right on 38 of 40 checked images. Errors in detecting the artifact or in deciding inside versus outside may bias the crossover in either direction; measuring that needs a check of the masks per trap and class, which we have not done yet.

Table 5: The four real-artifact cohorts (number of images, target class / other class). The real-artifact experiments use five grouped folds, so there is no fixed test split.

Cohort	Task	Artifact	No artifact	Trap A (inside)	Trap B (outside)
Skin lesions, ISIC 2019 (3 hospitals)	melanoma vs other	hair/ruler	157 / 706	942 / 2,655	969 / 4,412
Thyroid ultrasound, TN3K	malignant vs benign	calipers	749 / 1,052	292 / 858	99 / 227
Ovarian ultrasound, MMOTU	solid part vs cystic	calipers	178 / 170	276 / 341	103 / 91
Capsule endoscopy, SEE-AI	erosion vs polyp-like	debris, bubbles	343 / 164	412 / 404	1,009 / 231

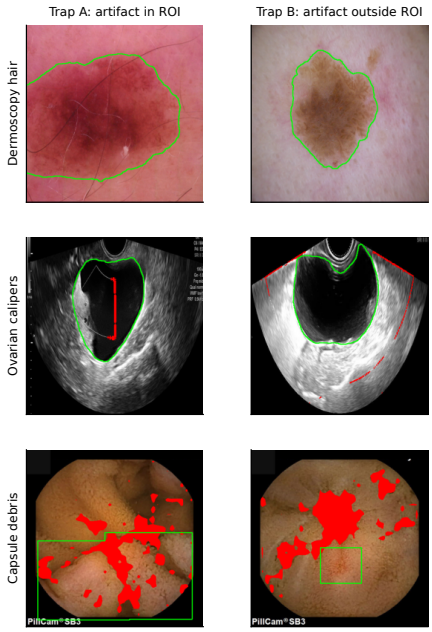


Figure 8: Real artifacts inside (Trap A) and outside (Trap B) the ROI. Green: ROI; red: detected artifact. ("Dermoscopy hair" = hair/ruler masks. Thyroid images are not shown because the data licence is unclear; the skin masks are not drawn for licence reasons.)

What we found The crossover was positive in all four cohorts with DINOv2 (Table 6). These four tests were planned in advance; in each of them, none of the 10,000 bootstrap replicates fell at or below zero. With the conservative two-sided formula $p = 2(k+1)/(B+1)$, where $k = 0$ replicates crossed zero and $B = 10{,}000$, this gives $p \approx 0.0002$, and $p \approx 0.0008$ after Holm correction for four tests; the same holds with one weight per lesion or near-duplicate group (see "Honest error bars"). It also held with MedSigLIP and with a convolutional network (ConvNeXt), so it is not specific to one type of model; it did not hold with DermLIP (Figure 9, Section 7).

Table 6: Real artifacts: effect of masking on the reversed test (masked − standard) in each trap, and the crossover.

Cohort	Encoder	Trap A (inside)	Trap B (outside)	Crossover (B − A)
Skin, hair/ruler	DINOv2	−0.049	+0.098	+0.147 [+0.111, +0.185]
Thyroid, calipers	DINOv2	+0.109	+0.348	+0.239 [+0.196, +0.280]
Ovary, calipers	DINOv2	+0.132	+0.302	+0.170 [+0.087, +0.252]
Capsule, debris	DINOv2	+0.084	+0.461	+0.377 [+0.327, +0.426]
Thyroid, calipers	MedSigLIP	+0.053	+0.248	+0.195 [+0.153, +0.238]
Ovary, calipers	MedSigLIP	+0.002	+0.180	+0.178 [+0.109, +0.248]
Capsule, debris	MedSigLIP	+0.117	+0.452	+0.335 [+0.286, +0.383]
Thyroid, calipers	ConvNeXt	−0.004	+0.271	+0.275 [+0.231, +0.320]
Capsule, debris	ConvNeXt	+0.084	+0.403	+0.320 [+0.268, +0.372]
Skin, hair/ruler	DermLIP	−0.104	−0.090	+0.014 [−0.037, +0.063]

How to read it For skin hair/ruler artifacts, masking *hurt* when they lay on the lesion. For calipers and debris, masking still helped in Trap A, because it removes other on-screen marks and clutter, but much less than in Trap B, and a large part of the shortcut stayed. In thyroid Trap A, the masked model scored 0.908 on the correlated test but only 0.398 on the reversed one; in Trap B masking closed this gap (0.785 vs 0.752). The general rule is: *masking helps much less when the artifact is inside the ROI*. Whether it actually hurts depends on how much useful image information masking also removes.

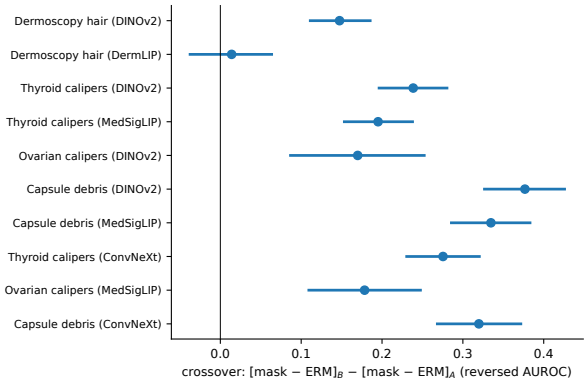


Figure 9: The crossover for every cohort and encoder (dots: estimate; lines: 95% interval). Right of zero = masking is worth more when the artifact is outside the ROI. "Dermoscopy hair" = hair/ruler masks.

7 Other things we learned

- **Methods that know which images carry the artifact help in both traps.** Reweighting the training images so that the artifact no longer predicts the label, or retraining the last layer on a balanced subset (DFR (Kirichenko et al., 2023)), helped in both traps of every cohort, wherever the artifact lay (Table 7). The catch: reweighting needs an artifact label for every training image, and DFR for every image of its balanced subset; clinical datasets rarely have such labels. Masking needs only the ROI.
- **Painting out the artifact can backfire.** In capsule images, painting over the detected debris *lowered* the score (−0.17 in Trap A). A likely reason is that erosions contain yellow-white material that looks like debris, so painting it out also removes disease evidence.
- **The hair/ruler shortcut is not only in the marked pixels.** Painting out the marked pixels closed less than half of the standard model's shortcut gap (47% in Trap A, 44% in Trap B). The rest may come from incomplete masks, from other artifacts, or from features that go with hair, such as age, sex and body site, which differ between the groups; these are hypotheses we have not tested.
- **DermLIP did not follow the rule.** With this dermatology encoder, masking hurt in *both* hair/ruler traps and there was no statistically clear crossover (+0.014 [−0.037, +0.063]). A likely reason is that it needs the skin around the lesion: masking lowered its clean-test AUROC as well.
- **Chest X-rays are a boundary case.** Real chest drains lie inside the lungs and are an extreme shortcut (standard model, RAD-DINO: reversed AUROC 0.19 against 0.89 on the clean test). Lung masking barely helped (0.23), while DFR reached 0.67. With real devices (catheters in the lungs vs breathing tubes outside them), RAD-DINO showed no crossover and MedSigLIP a positive one in all four findings tested. The two device groups also differ in which other devices are present, so we do not count chest X-rays as a confirmation.

Table 7: Masking against methods that use artifact labels (reversed test, change vs the standard model, DINOv2).

	Skin, hair/ruler		Thyroid, calipers		Capsule, debris	
Method	Trap A	Trap B	Trap A	Trap B	Trap A	Trap B
ROI masking (needs ROI only)	−0.049	+0.098	+0.109	+0.348	+0.084	+0.461
Balanced reweighting (needs labels)	+0.213	+0.204	+0.388	+0.254	+0.179	+0.244
DFR (needs labels)	+0.196	+0.196	+0.463	+0.312	+0.234	+0.287
Painting out the artifact (needs its mask)	+0.098	+0.100	+0.102	+0.039	−0.172	−0.115

8 Where we are

Table 8: What the work so far shows.

Question	Answer	Evidence
Does masking depend on where the artifact sits?	Yes: it helps when the artifact is outside the ROI and loses its benefit (often turns harmful) when it is inside	Steps 1 and 3: 10 of 11 experiments; both ruler sizes
Why?	The artifact inside survives, its competitors are deleted, and the model relies on it more	Step 2: four checks
Does it hold for real artifacts?	Yes, in all four cohorts with DINOv2, and in every cohort tested with MedSigLIP (3) and ConvNeXt (2)	Step 4: four planned tests, all significant
Exceptions	DermLIP (likely needs the skin around the lesion); RAD-DINO (does not use a tube outside the lungs)	Steps 3 and 4

Questions for discussion

1. Is the reversed-test AUROC acceptable as the main score, or should a clinical measure (sensitivity at a fixed specificity) be added from the start?
2. Should the rule be stated as "masking is worth less inside the ROI" (true in all real cohorts) rather than "masking hurts inside the ROI" (true for skin hair/ruler artifacts and in 7 of 11 moving-artifact experiments)?
3. The artifact masks were checked by eye by a non-clinician. Should a clinician check a sample?
4. How should chest X-rays be presented: as a boundary case, or left out of the main text?

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